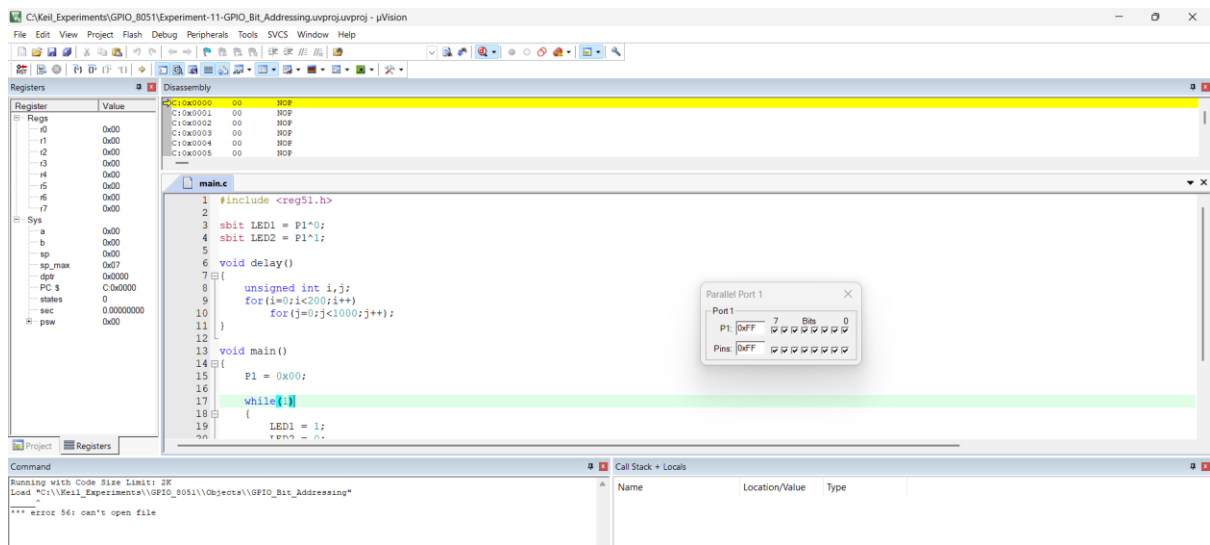


EXP-11: STUDY OF GPIO & BIT ADDRESSING IN AT89C51



1) Source Code

```
#include <reg51.h>
sbit LED1 = P1^0;
sbit LED2 = P1^1;
void delay()
{
    unsigned int i,j;
    for(i=0;i<200;i++)
    for(j=0;j<1000;j++);
}
void main()
{
    while(1)
    {
        LED1 = 1;
        LED2 = 0;
        delay();
        LED1 = 0;
        LED2 = 1;
        delay();
    }
}
```

Output:

Registers

Register	Value
r0	0x00
r1	0x00
r2	0x00
r3	0x00
r4	0x00
r5	0x00
r6	0x00
r7	0x00
a	0x00
b	0x00
sp	0x07
sp_max	0x07
dpr	0x0000
PC	0x0822
status	392
sec	0.00019600
psw	0x00

Disassembly

```
20: LED2 = 0;
21: CLR LED2(0x90,1)
22: delay();
23: LCALL delay(C:0800)
24: LED1 = 0;
```

main.c

```
STARTUPAS1
9: for(i=0;i<200;i++)
10: for(j=0;j<1000;j++);
11: }
12: void main()
13: {
14: P1 = 0x00;
15: while(1)
16: {
17: LED1 = 1;
18: LED2 = 0;
19: delay();
20: LED1 = 0;
21: LED2 = 1;
22: delay();
23: }
24: }
```

Parallel Port 1

Port	7	6	5	4	3	2	1	0
P1	0x01							
Pins	0x01							

Command

Running with Code Size Limit: 2K
Load "C:\Keil_Experiments\GPIO_8051\Objects\GPIO_Bit_Addresssing"

Call Stack + Locals

Name	Location/Value	Type
MAIN	C:0x081D	

Simulation t1: 0.00019600 sec L20 C:1 CAP.NUM: SCRL OVR: R/W

Registers

Register	Value
r0	0x00
r1	0x00
r2	0x00
r3	0x00
r4	0x00
r5	0x08
r6	0x08
r7	0x08
a	0x00
b	0x00
sp	0x07
sp_max	0x09
dpr	0x0000
PC	0x0828
status	1217667502
sec	608.83375100
psw	0x00

Disassembly

```
24: LED2 = 1;
25: SETB LED2(0x90,1)
26: delay();
27: LCALL delay(C:0800)
28: SJMP C:0820
```

main.c

```
STARTUPAS1
9: for(i=0;i<200;i++)
10: for(j=0;j<1000;j++);
11: }
12: void main()
13: {
14: P1 = 0x00;
15: while(1)
16: {
17: LED1 = 1;
18: LED2 = 0;
19: delay();
20: LED1 = 0;
21: LED2 = 1;
22: delay();
23: }
24: }
```

Parallel Port 1

Port	7	6	5	4	3	2	1	0
P1	0x02							
Pins	0x02							

Command

Running with Code Size Limit: 2K
Load "C:\Keil_Experiments\GPIO_8051\Objects\GPIO_Bit_Addresssing"

Call Stack + Locals

Name	Location/Value	Type
MAIN	C:0x081D	

Simulation t1: 0.00019600 sec L20 C:1 CAP.NUM: SCRL OVR: R/W